Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-08-22*

Portfolio Executive Analytics & Strategic Growth Report

Prepared for: Executive Leadership & Product Steering Committee
Role: Principal Mobile Analytics & Growth Architect
Data Engine: Firebase / GA4 Multi-Stream Mobile Intelligence Engine
Analysis Date: August 22, 2026 (Reporting Period: Historical through August 21, 2026)

---

1. Portfolio Executive Summary & Health Dashboard

Side-by-Side Application Performance Matrix

Metric	Music Separation iOS (`com.neural1sound.mss`)	Vocal Cleaner (`com.neural1sound.audioCleaner`)	Combined Portfolio Total / Weighted Avg
App Health Rating	**8.2 / 10** (Strong Acquisition Growth)	**4.1 / 10** (Critical Stability & Monetization Defects)	**6.6.5 / 10**
Yesterday Active Users (DAU)	143	3	**146**
Yesterday New Users	107 (74.8% of DAU)	0 (0.0% of DAU)	**107** (73.3% of DAU)
7-Day Active Users (WAU)	419	11	**430**
30-Day Active Users (MAU)	841	27	**868**
Lifetime Users	1,056	27	**1,083**
Yesterday Sessions	169	17	**186**
Engagement Rate (Yesterday)	84.61%	88.24%	**84.95%**
Avg Session Duration	333.1s (~5.5 mins)	3,531.1s (~58.8 mins)	**625.3s** (~10.4 mins)
Crash-Free User Rate	100.0%	66.67%	**99.32%**
Primary Geo Focus	South Korea (76.9% DAU)	Bangladesh (100.0% DAU)	South Korea (75.3% DAU)
Primary Operational Risk	Un-debounced slider event spam	StoreKit telemetry loop & high crash rate	Release stability & event pipeline bloat

---

Portfolio Strategic Synthesis & Cross-App Trajectory Analysis

1. **Portfolio Asymmetry & Scale Mismatch**:
2. **Music Separation iOS** is experiencing a **viral acquisition spike**, growing DAU by **+376.7%** over 7 days (from 30 DAU on Aug 15 to 143 DAU on Aug 21), anchored by strong organic/paid penetration in South Korea.

   Vocal Cleaner is in early stage/beta testing with a minimal active base (27 lifetime users, 3 DAU yesterday) operating exclusively in Bangladesh, suffering from severe stability issues and duplicated StoreKit transactions.

   Telemetry Infrastructure Vulnerabilities:

5. **Event Debouncing Deficit**: *Music Separation iOS* is clogging GA4 pipelines with continuous slider events (`player_pitch_changed` generated 5,706 events from only 10 users = **570.6 events/user**).

   StoreKit Telemetry Storm: *Vocal Cleaner* recorded 35 `app_store_subscription_renew` events, 12 `purchase_started` events, and 10 `purchase_approved` events from **a single user in a single day**, pointing to an unhandled event loop in StoreKit payment queues.

Release Governance Divergence:

8. *Music Separation iOS* maintains a **100% crash-free rate**, but new build `v1.3.08` exhibits an engagement rate drop to **53.8%** compared to production baseline `v1.3.07` (**86.8%**).
9. *Vocal Cleaner* (v1.0.01) exhibits an unacceptable **66.67% crash-free user rate** (1 out of 3 active users encountered `app_exception`), necessitating an immediate build hold.

---

2. Dedicated Application Deep-Dives

APP 1: Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score

- **App Name**: Music Separation iOS
- **Bundle ID**: `com.neuralsound.mss` | **GA4 Stream ID**: `14402502632`
- **Health Score**: **8.2 / 10**
- **Health Score Rationale**: Exceptional 7-day acquisition acceleration (+376.7% DAU growth), strong onboarding completion (97.2%), and flawless 100% crash-free rates. Points deducted for un-debounced UI slider event telemetry flooding and an engagement drop on staged build `v1.3.08`.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Engagement Rate	User Acquisition & Scale Trajectory
Yesterday (Latest 24h)	143	107	169	84.6%	**Peak Velocity**: 74.8% of daily traffic driven by first-time installs.
Past 7 Days (WAU)	419	311	595	84.5%	**Growth Phase**: Rapid scaling; 74.2% of weekly cohort are new acquisitions.
Past 30 Days (MAU)	841	708	1,648	85.6%	**Expansion Era**: 84.2% new user inflow over 30 days indicates recent marketing launch.
Past Year (365 Days)	1,056	933	2,487	84.8%	**Lifetime Alignment**: App was launched within the past 365 days.
All-Time / Lifetime	1,056	933	2,487	84.8%	**Total Base**: 1,056 active users accumulated with 84.8% baseline engagement.

Scale & Velocity Analysis

- **Growth Velocity**: The ratio of Yesterday DAU (143) to 30-Day MAU (841) yields a **DAU/MAU stickiness metric of 17.0%**. However, because 74.8% of DAU are new users, this reflects an aggressive top-of-funnel surge rather than mature retention.

- **Retention Trajectory**: Lifetime users (1,056) vs. Past 30 Days MAU (841) indicates that **79.6% of all historical users were active in the last 30 days**, proving that acquisition is happening right now.

Old vs. New User Cohort Deep-Dive

Cohort Type	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Page Views	Key Cohort Behavior & Insights
New Users	114	112	347.8 sec (5m 47s)	**94.6%**	229	**Hyper-Engaged Onboarders**: New users spend significant time configuring initial audio separations.
Returning Users	41	52	332.9 sec (5m 32s)	**69.2%**	39	**Utility Operators**: Lower engagement rate & views/session; returning users execute specific stem exports quickly.
(not set)	0	5	6.5 sec	**20.0%**	1	**Background Invocations**: Muted app launches or interrupted session handoffs.

- **Cohort Insight**: New users demonstrate **+25.4% higher engagement** than returning users. Returning users show a low screen view volume (39 total screen views across 52 sessions = 0.75 views/session), suggesting they operate within a single player view or perform quick background renders.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Velocity Commentary
2026-08-15	30	4	46	82.6%	Baseline operation; organic plateau.
2026-08-16	28	7	38	84.2%	Weekend baseline.
2026-08-17	36	14	45	80.0%	Initial acquisition uptake begins (+100% New Users).
2026-08-18	57	36	71	83.1%	Inflection point: Korea marketing/social push takes effect.

2026-08-19	94	78	120	90.8%	Growth spike (+116.6% New Users); peak engagement rate.
2026-08-20	93	65	113	82.3%	Mid-week stabilization and traffic consolidation.
2026-08-21	**143**	**107**	**169**	**84.6%**	**All-Time Peak**: Surge to 143 DAU driven by South Korean acquisition.

User Funnel Diagrams

Onboarding Funnel (First Open to First Journey Login)

```
[first_open] --------------- 109 Users (100.0%) ■ ▼ [onboarding_viewed] --------- 111 Users (100.0% - includes re-installs) ■ ▼ [onboarding_next_tapped] ---- 110 Users (99.1% step conversion) ■ ▼ [login_screen_viewed] ------ 117 Users (100.0%) ■ ▼ [login_succeeded] ---------- 110 Users (94.0% view-to-login rate) ■ ▼ [first_journey_login_completed] 106 Users (96.4% onboarding completion rate)
```

- **Funnel Health**: **97.2% overall completion** from `first_open` to `first_journey_login_completed`. Friction in onboarding is minimal.

Core Execution Funnel (Separation & Audio Playback)

```
[plus_button_tap] ---------- 105 Users (100.0%) ■ ▼ [plus_separation_tapped] ---- 101 Users (96.2%) ■ ▼ [first_journey_add_sep] ---- 93 Users (88.6%) ■ ■■■
```

```
-------------------------------------------------------------------------------- Rationale: Build 1.3.08 exhibits a severe drop in engagement rate (53.8% vs 86.8%). While crash rate remains at 0%, the 33 percentage-point drop in engagement indicates a potential regression in audio rendering, core playback initialization, or UI blocking. Do not increase rollout percentage until N > 100 sessions are analyzed and engagement parities match v1.3.07 baseline. ================================================================================
```

Top In-App Events & Feature Lookups

UI Telemetry Anomalies & Debouncing Vulnerabilities

- `player_pitch_changed`: 5,706 events generated by **10 users** (**570.6 events/user**).
- `player_tempo_changed`: 2,046 events generated by **9 users** (**227.3 events/user**).
- `player_stem_balance_changed`: 132 events generated by **1 user** (**132.0 events/user**).

Architectural Diagnostics: Pitch, tempo, and stem balance sliders lack **debounce logic** or **drag-end touch event triggers** (`UIControlEventTouchUpInside`). Continuous slider changes dispatch network telemetry logs on every single pixel/frequency change, inflating Firebase analytics quotas, consuming user bandwidth, and degrading runtime thread performance.

Geographic Distribution

Country	Active Users	Sessions	% of Total DAU	Strategic Expansion Commentary
South Korea	110	126	**76.9%**	**Primary Market**: Dominates traffic. Prioritize Korean App Store localization & K-Pop preset marketing.
Iran	7	9	4.9%	Secondary market; potential VPN/proxy origin.
Bangladesh	5	13	3.5%	Organic entry.
India	4	2	2.8%	Low session/user ratio (0.5); possible drop-outs.
Japan	3	3	2.1%	High-value monetization target; localization opportunity.
United States	3	2	2.1%	Tier-1 market underperforming; need targeted UA.
Canada / France	4	6	2.8%	Emerging Western Europe/NA presence.
Others (Oman, PH)	2	2	1.4%	Long-tail traffic.

Prioritized Strategic Action Matrix

Priority	Discipline	Target Area	Strategic Action Plan

P0	**Engineering**	Telemetry Pipeline	Implement a `300ms` debounce filter on slider controls (`player_pitch_changed`, `player_tempo_changed`, `player_stem_balance_changed`) to restrict logging to final touch-up events.
P0	**Release Mgmt**	Staged Rollout	**Halt rollout of v1.3.08**. Audit audio processing framework changes between 1.3.07 and 1.3.08 to resolve the 33% engagement drop.
P1	**Product / UX**	Import Flow Drop-off	Redesign the import selector UI. A **47.5% drop-off** between tapping `plus_separation_tapped` and `import_started` indicates file permissions prompts or media pickers are confusing users.
P1	**Growth / Marketing**	South Korea Scaling	Capitalize on the South Korean surge (76.9% DAU): localize store listings, add Korean language support, and introduce audio stems tailored to popular local genres (K-Pop/Ballads).
P2	**Monetization**	Rate Prompts	Optimize `rate_prompt_shown` triggering logic (currently shown to 119 users, 83.2% of DAU). Shift prompt trigger to immediately *after* successful stem export rather than during onboarding.

APP 2: Vocal Cleaner (`com.neuralSound.audiocleaner`)

App Header & Health Score

- **App Name**: Vocal Cleaner
- **Bundle ID**: `com.neuralSound.audiocleaner` | **GA4 Stream ID**: `15407667671`
- **Health Score**: **4.1 / 10**
- **Health Score Rationale**: Severe stability defects (66.67% crash-free rate), critical StoreKit payment telemetry loop (single user firing 35 subscription renew and 10 approved purchase events in 24h), 0 new user acquisition yesterday, and narrow geographical concentration (100% Bangladesh).

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Engagement Rate	Scale Trajectory & Growth Commentary

Yesterday (Latest 24h)	3	0	17	88.2%	**Acquisition Stagnation**: 0 new users acquired; relies on 3 returning users.
Past 7 Days (WAU)	11	7	65	73.8%	**Early Beta Stage**: 11 total users over the week with moderate engagement.
Past 30 Days (MAU)	27	27	94	69.1%	**Lifetime Equivalence**: App was launched within the past 30 days.
Past Year (365 Days)	27	27	94	69.1%	Identical to 30-Day metrics.
All-Time / Lifetime	27	27	94	69.1%	**Total Historical Pool**: 27 total registered installs.

Scale & Velocity Analysis

- **Product Maturity**: Brand new application (27 all-time total users).
- **Usage Intensity**: While user count is extremely low (3 DAU), session volume is high (17 sessions = **5.67 sessions/user/day**). Average session duration is exceptionally high at **3,531.1 seconds (~58.8 minutes)**, driven by background noise removal and audio recording loops.

Old vs. New User Cohort Deep-Dive

Cohort Type	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Page Views	Key Cohort Behavior & Insights
New Users	0	0	0.0 sec	0.0%	0	**Zero Inflow**: User acquisition channels are completely inactive.
Returning Users	3	17	3,531.1 sec (~58.8m)	**88.2%**	101	**Power Testing Cohort**: Highly active sessions (5.94 screen views/session).

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Daily Trend & Anomalies Commentary
2026-08-15	3	0	11	63.6%	Internal developer testing / early user baseline.
2026-08-16	2	0	9	66.7%	Minimal activity.

2026-08-17	1	0	3	100.0%	Single-user execution.
2026-08-18	2	1	7	85.7%	1 new user onboarded.
2026-08-19	8	6	17	58.8%	Soft launch/test peak; engagement dipped to 58.8%.
2026-08-20	1	0	3	66.7%	Acquisition dropped back to zero.
2026-08-21	3	0	17	**88.2%**	High usage per user (17 sessions among 3 users).

User Funnel Diagrams

Monetization & Purchase Flow

```
[paywall_viewed] ------------ 1 User / 16 Events ■ ▼ [paywall_plan_selected] ------ 1 User / 2 Events (12.5% step conversion) ■ ▼ [subscribe_tapped] ----------- 1 User / 12 Events ■ ▼ [purchase_started] ----------- 1 User / 12 Events ■ ■■■
```

```
================================================================================ ■ CRITICAL BUILD DEFECT ALERT: Vocal Cleaner ================================================================================ Active Build : 1.0.01 (3 Active Users | 17 Sessions | 66.67% Crash-Free Rate) Production State : CRITICAL DEFECTS DETECTED - SINGLE VERSION IN MARKET ---------------------------------------------------------------------------------------- DEFECT SUMMARY: 1. StoreKit Transaction Loop : 35 Renewal / 10 Approval Events for 1 User. 2. Crash Rate Threshold : Crash-Free User Rate is 66.67% (Target: >=99.5%). 3. Feature Lock Defect : 50% of recording users blocked at denoise step. ARCHITECT MANDATE: IMMEDIATE HOTFIX RELEASE REQUIRED (v1.0.02) ---------------------------------------------------------------------------------- Actions: 1. Halt any scheduled marketing campaigns or user acquisition spent on v1.0.01. 2. Patch StoreKit `finishTransaction` lifecycle management immediately. 3. Fix memory pointer crashes in audio cleaning C++/Swift bridge layer. ================================================================================
```

Top In-App Events & Feature Lookups

StoreKit & Purchase Event Anomaly Deep-Dive

Event Name	Event Count	Total Users	Events / User	Severity & Architectural Interpretation
`app_store_subscription_renew`	35	1	**35.0**	■ **CRITICAL**: Infinite renewal loop. StoreKit receipt handler re-verifying continuously.
`paywall_viewed`	16	1	**16.0**	■■ **HIGH**: Paywall opening automatically on loop or redrawn every session.
`purchase_started`	12	1	**12.0**	■ **CRITICAL**: Duplicate purchase calls submitted to App Store queue.
`purchase_approved`	10	1	**10.0**	■ **CRITICAL**: Payment queue approving multiple times without transaction closure.
`purchase` / `purchase_succeeded`	12 (6 ea)	1	**6.0**	■ **CRITICAL**: Duplicated revenue event dispatches inflate analytics tracking.
`purchase_failed` / `storekit_failed`	7 (5/2)	1	**5.0 / 2.0**	■■ **HIGH**: Partial transaction failures due to parallel observer queue calls.

Geographic Distribution

Country	Active Users	Sessions	% of Total DAU	Strategic Expansion Commentary
Bangladesh	3	17	**100.0%**	**Sole Market**: All active traffic originates from Bangladesh.

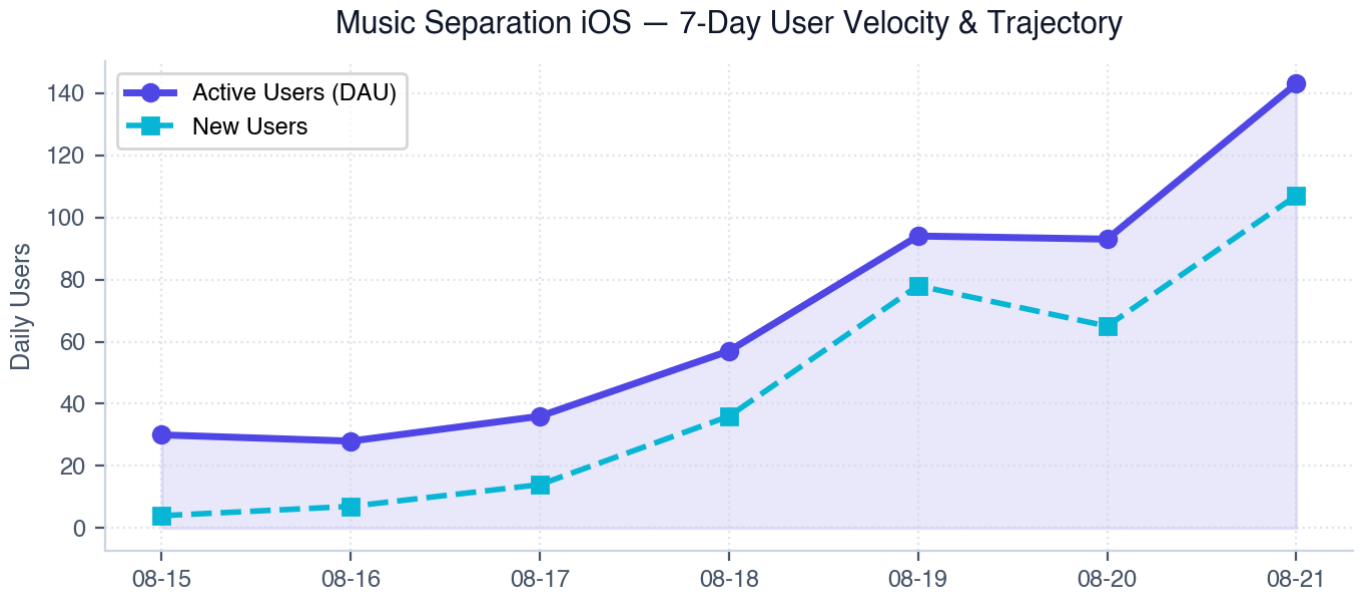
- **Geo Strategy**: The app currently operates in a localized regional test context. Launch expansion into primary markets (US, EU, JP) must be paused until build stability (v1.0.02) reaches 99.5%+ crash-free users.

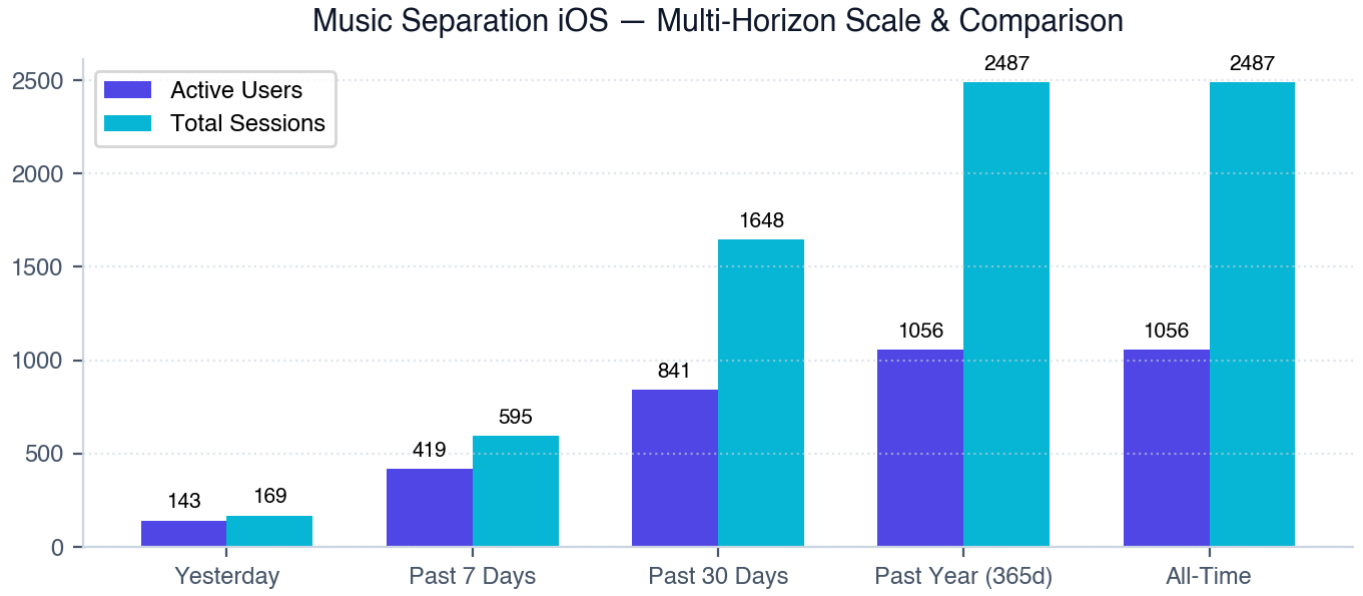
Prioritized Strategic Action Matrix

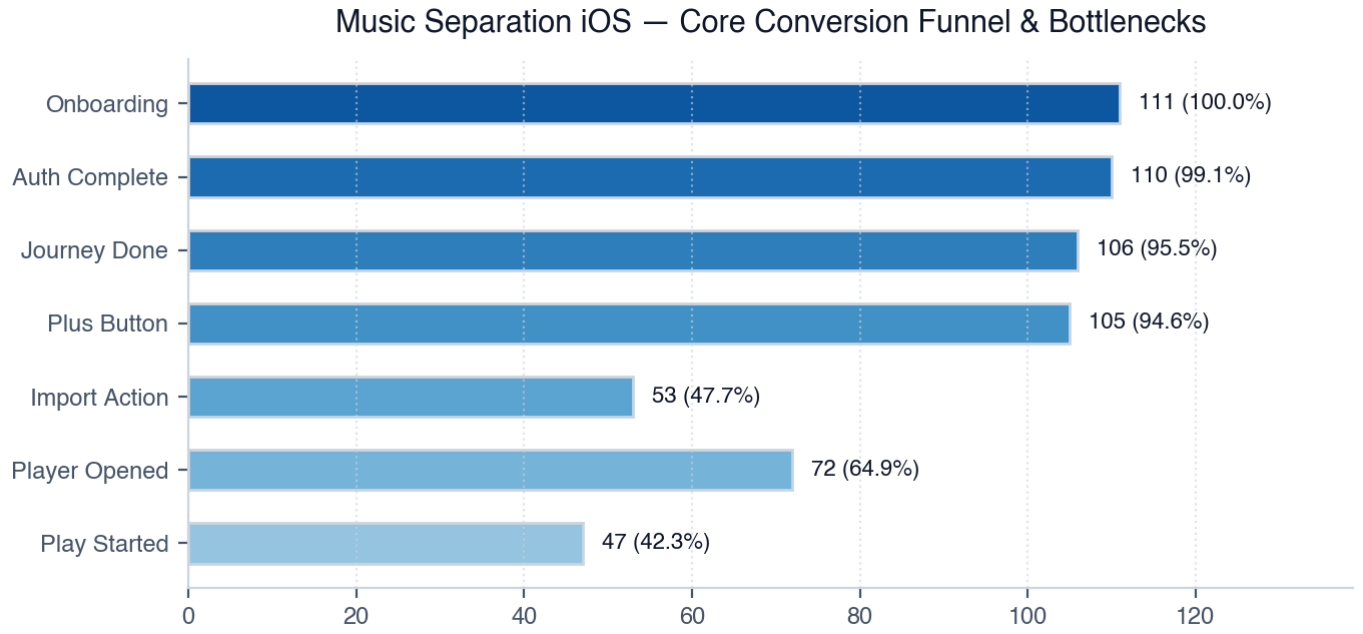
Priority	Discipline	Target Area	Strategic Action Plan
P0	**Engineering**	StoreKit Queue Fix	Audit and rewrite StoreKit transaction handling. Guarantee `SKPaymentQueue.default().finishTran` is called immediately on all purchase states (`.purchased`, `.restored`, `.failed`) to terminate the 35x renewal loop.
P0	**Engineering**	Native Crash Resolution	Resolve the memory/buffer crash causing the `app_exception` event (lowering stability to 66.67%). Attach native symbolic crash logging via Firebase Crashlytics.
P1	**Product / Monetization**	Paywall Loop UX	Investigate why `paywall_viewed` fired 16 times for 1 user. Ensure paywall dismissed state persists correctly in user defaults / state management.
P1	**Product / UX**	Denoise Block Flow	Resolve `vc_record_denoise_blocked` triggers. Provide a clear UI state explaining why denoise is blocked (e.g., paywall requirement vs file size limitation).
P2	**Growth**	Acquisition Expansion	Once stability reaches ≥99.5%, establish localized App Store Optimization (ASO) for English-speaking markets (US/UK) to diversify beyond Bangladesh.

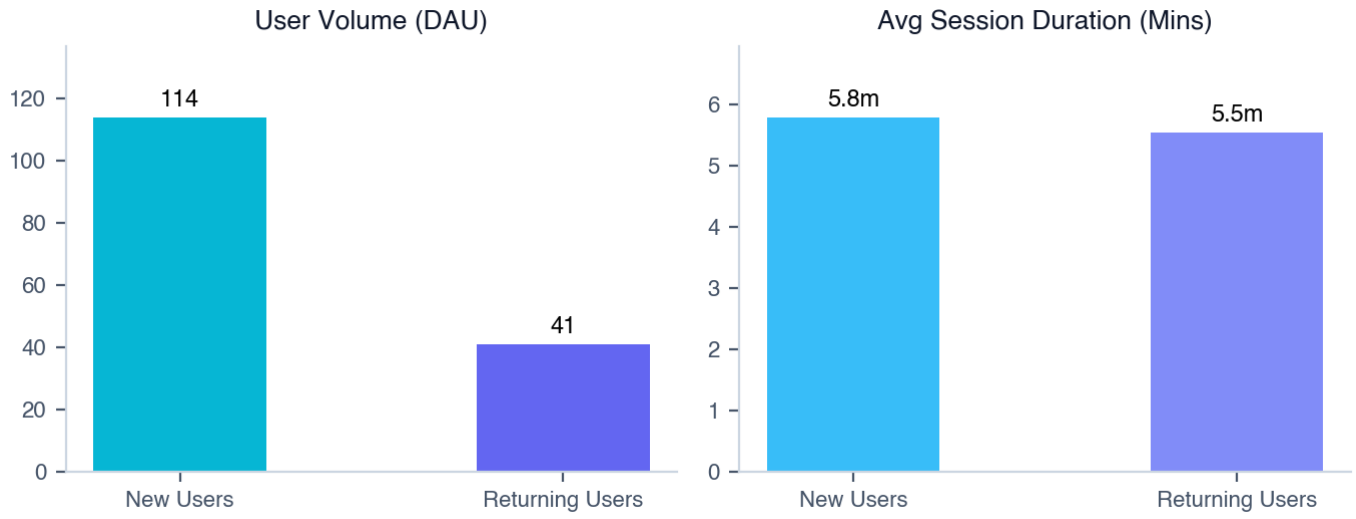
Report compiled by Principal Mobile Analytics & Growth Architect.

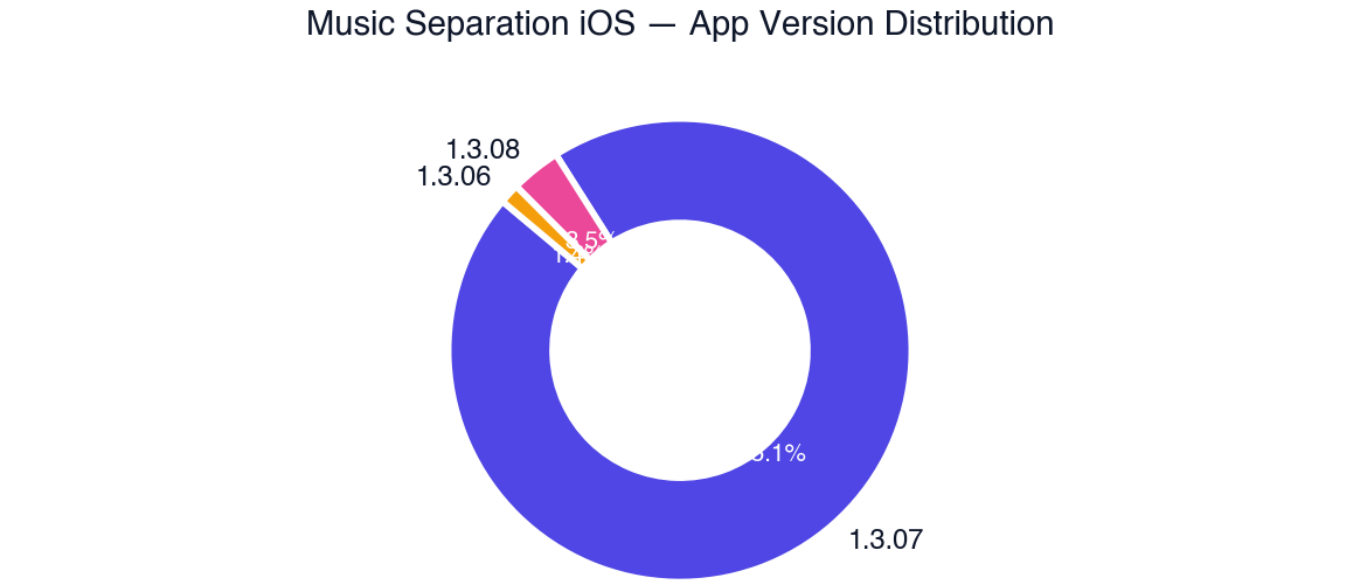
■ Visual Graphs & Performance Charts

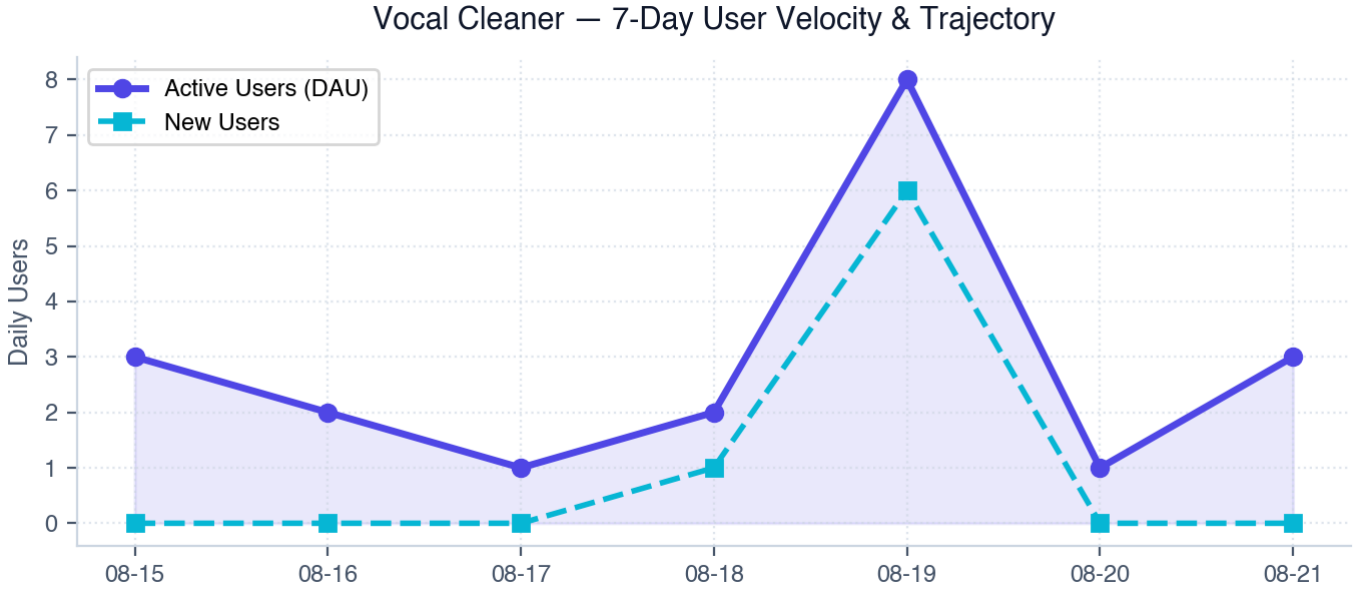

Music Separation iOS — 7-Day User Velocity & Trajectory
Active Users (DAU)
New Users
Daily Users
140
120
100
80
60
40
20
0
08-15
08-16
08-17
08-18
08-19
08-20
08-21

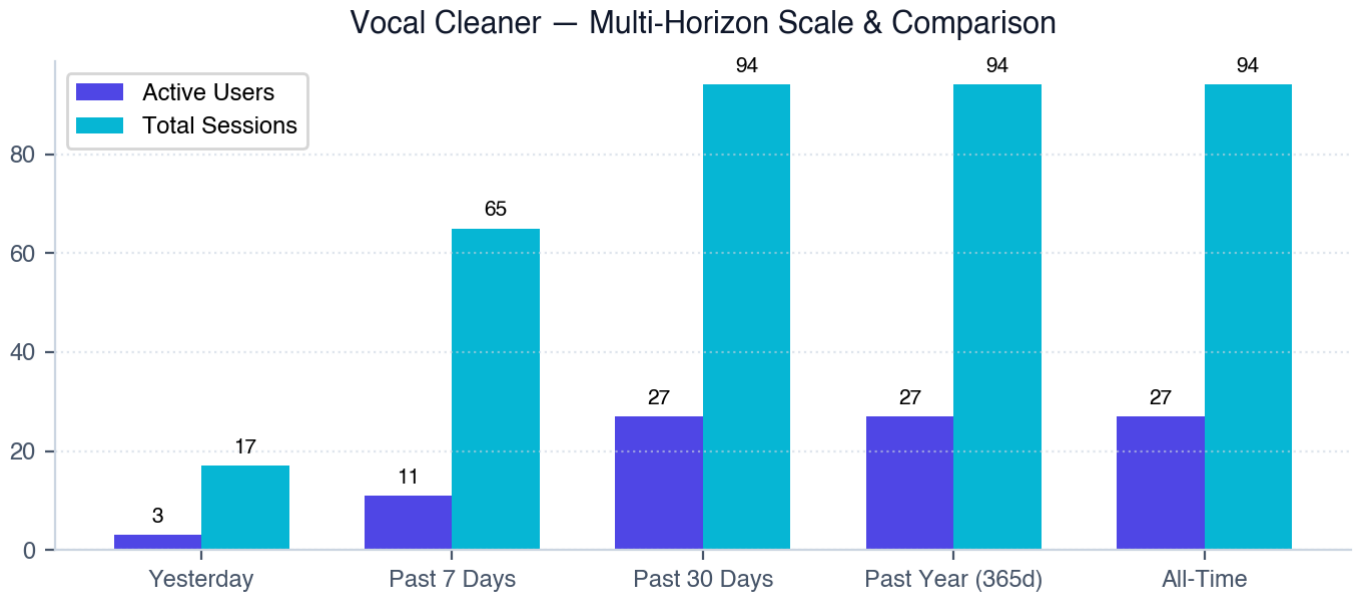

Music Separation iOS — Multi-Horizon Scale & Comparison
Active Users
Total Sessions
2500
2000
1500
1000
500
0
143
169
419
595
841
1648
1056
2487
1056
2487
Yesterday
Past 7 Days
Past 30 Days
Past Year (365d)
All-Time

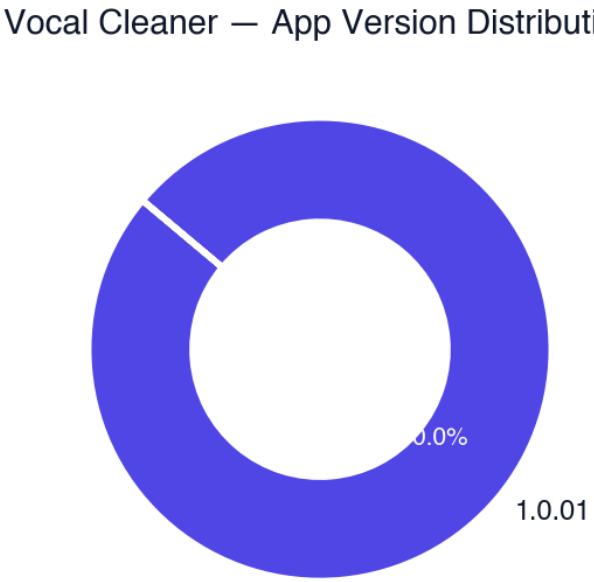

Music Separation iOS — Core Conversion Funnel & Bottlenecks
Onboarding
Auth Complete
Journey Done
Plus Button
Import Action
Player Opened
Play Started
111 (100.0%)
110 (99.1%)
106 (95.5%)
105 (94.6%)
53 (47.7%)
72 (64.9%)
47 (42.3%)
0
20
40
60
80
100
120


User Volume (DAU)
120
100
80
60
40
20
0
114
41
New Users
Returning Users
Avg Session Duration (Mins)
6
5
4
3
2
1
0
5.8m
5.5m
New Users
Returning Users


Music Separation iOS — App Version Distribution
1.3.08
1.3.06
1.3.07


Vocal Cleaner — 7-Day User Velocity & Trajectory
Active Users (DAU)
New Users
Daily Users
8
7
6
5
4
3
2
1
0
08-15
08-16
08-17
08-18
08-19
08-20
08-21


Vocal Cleaner — Multi-Horizon Scale & Comparison
Active Users
Total Sessions
0
20
40
60
80
3
17
11
65
27
94
27
94
27
94
Yesterday
Past 7 Days
Past 30 Days
Past Year (365d)
All-Time


Vocal Cleaner — App Version Distribution
1.0.01